

- **Natural language processing (NLP):** It can perform various NLP tasks, processing human language and allowing users to ask questions conversationally.
- **Answering questions:** Generative AI can answer questions based on a knowledge base.
- **Personalisation:** It can be fine-tuned for specific use cases.
- **Multi-language support:** Generative AI can handle NLP tasks in multiple languages.
- **Advanced semantic search:** It enables semantic search in large volumes of structured and unstructured data, including databases, documents, etc. It also supports auto data indexing using custom AI/ML based data processing for efficient embedding.
- **Content moderation:** It has smart content filtering and moderation engines for query and response, trainable on enterprise-specific data.
- **Integration:** It can integrate with various applications and data sources within an organisation, including CRM, ERP, and other proprietary systems, to access and analyse data via APIs.
- **Role-based access control:** The system can be configured with role-based access control, ensuring that users only have authorised access to data.
- **Software coding:** Generative AI can generate, translate, explain and verify code.

Real-world use cases of generative AI

Generative AI's use cases are vast and continually evolving. Businesses across industries are experimenting with different ways to incorporate generative AI to enhance efficiency and decision-making capabilities. Generative AI applications improve experiences, reduce costs and increase revenues for enterprises.

Here is a summary of the generative AI use cases across various industries.

Healthcare and pharma

Generative AI-based applications help healthcare professionals be more productive by identifying potential issues upfront, providing insights to deliver interconnected healthcare and improve the health of patients. It helps in:

Better customer experience:

Automates administrative tasks such as processing claims, scheduling appointments, and managing medical records.

Patient health summary: Helps provide healthcare decision support by generating personalised patient health summaries, speeding up patient response times, and improving the patient experience.

Faster analysis of publications:

Generative AI helps in reducing the time it takes to create research publications on specific drugs by analysing vast amounts of data from multiple sources faster than ever.

Personalised medicine: Generative AI-based individualised treatment plans are tailored to a patient's genetic makeup, medical history, life style, etc.

Healthcare virtual assistant: It provides end users with conversational and engaging access to the most relevant and accurate healthcare services and information.

Manufacturing

Generative AI enables manufacturers to make better use of their data, leading to advancements in predictive maintenance and demand forecasting. It also helps in simulating manufacturing quality, improving production speed, and material efficiency.

Predictive maintenance: It helps in estimating the life of machines and their components. Generative AI can provide proactive information to technicians about repairs and replacement of parts and machines, thus reducing downtime.

Performance efficiency:

Anticipating the problems proactively can cover production disruptions,

bottlenecks, and safety risks in real time.

Other uses in the manufacturing industry include:

- Yield, energy and throughput optimisation
- Digital simulations
- Sales and demand forecasting
- Logistics network optimisation

Retail

Generative AI in retail helps in personalising offerings, brand management, as well as optimising marketing and sales activities.

Personalised offerings: It enables retailers to deliver customised experiences, offerings and pricing, and plan according to customer demand, modernising the online and physical buying experience.

Dynamic pricing and planning:

It helps predict demand for different products, enabling more confident pricing and stocking decisions.

Other uses include:

- Campaign management
- Content management
- Augmented customer support
- Search engine optimisation

Banking

Generative AI applications help in delivering a personalised banking experience to customers. There are other benefits too.

Risk mitigation and portfolio optimisation: Generative AI helps banks to build data foundations for developing risk models, and identify events that are impacting the bank. It can help mitigate risks to optimise portfolios.

Customer pattern analysis:

Generative AI can analyse patterns in banking data at scale, helping relationship managers and customer representatives to identify customer preferences, anticipate needs, and create personalised banking experiences.

Customer financial planning:

Generative AI can be used to automate customer service, identify trends in customer behaviour, and predict customer